

Brief Report

What include

- Docker Segmentation Process and Implementation
- Applicability Compliance Lists
- Time Planning Table

1. Docker Segmentation Process and Implementation

Containers / log-server

<



log-server

 4135b9a2f4f4

 alpine:3.23

Logs

Inspect

Bind mounts

Exec

Files

Stats

(1/4) Installing ncurses-terminfo-base (6.5_p20251123-r0)

(2/4) Installing libncursesw (6.5_p20251123-r0)

(3/4) Installing readline (8.3.1-r0)

(4/4) Installing socat (1.8.0.3-r1)

Executing busybox-1.37.0-r30.trigger

OK: 9464 KiB in 20 packages

2026/06/06 17:24:53 socat[1] W address is opened in read-write mode but only supports read-only

2026/06/06 17:24:53 socat[1] W address is opened in read-write mode but only supports write-only

test log event

> 2026/06/06 17:38:12.000474253 length=15 from=0 to=14

test log event

(Log Forwarding Testing : Simplex Mode)

<



s2

D:\Workspaces\Test\S2

View configurations







 log-server

alpine:3.23







 db

postgres:18.4







 kms

alpine:3.23







 pos

alpine:3.23







 app-server

alpine:3.23







 internal-fw

s2-internal-fw







 perimeter-fw

s2-perimeter-fw







stderr: warning: enabling 'strict' segmentation for 'core' connections

initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.

waiting for server to start....2026-06-06 17:24:53.813 UTC [54] LOG: starting PostgreSQL 18.4 (Debian 18.4-1.pgdb13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit

2026-06-06 17:24:53.820 UTC [54] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"

2026-06-06 17:24:53.868 UTC [60] LOG: database system was shut down at 2026-06-06 17:24:53 UTC

2026-06-06 17:24:53.879 UTC [54] LOG: database system is ready to accept connections

done

server started

CREATE DATABASE

/usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*

waiting for server to shut down....2026-06-06 17:24:54.235 UTC [54] LOG: received fast shutdown request

2026-06-06 17:24:54.242 UTC [54] LOG: aborting any active transactions

2026-06-06 17:24:54.246 UTC [54] LOG: background worker "logical replication launcher" (PID 63) exited with exit code 1

2026-06-06 17:24:54.246 UTC [58] LOG: shutting down

2026-06-06 17:24:54.254 UTC [58] LOG: checkpoint starting: shutdown immediate

2026-06-06 17:24:54.412 UTC [58] LOG: checkpoint complete: wrote 943 buffers (5.8%), wrote 3 SLRU buffers; 0 WAL file(s) added, 0 removed, 0 recycled; write=0.840 s, sync=0.093 s, total=0.166 s; sync files=303, longest=0.029 s, average=0.001 s; distance=4352 kB, estimate=4352 kB; lsn=0/1B9FBA8, redo lsn=0/1B9FBA8

2026-06-06 17:24:54.440 UTC [54] LOG: database system is shut down

done

server stopped

PostgreSQL init process complete; ready for start up.

2026-06-06 17:24:54.586 UTC [1] LOG: starting PostgreSQL 18.4 (Debian 18.4-1.pgdb13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit

2026-06-06 17:24:54.587 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432

2026-06-06 17:24:54.587 UTC [1] LOG: listening on IPv6 address ":::", port 5432

2026-06-06 17:24:54.602 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"

2026-06-06 17:24:54.622 UTC [76] LOG: database system was shut down at 2026-06-06 17:24:54 UTC

2026-06-06 17:24:54.633 UTC [1] LOG: database system is ready to accept connections

2026-06-06 17:29:54.705 UTC [74] LOG: checkpoint starting: time

2026-06-06 17:29:59.121 UTC [74] LOG: checkpoint complete: wrote 43 buffers (0.3%), wrote 3 SLRU buffers; 0 WAL file(s) added, 0 removed, 0 recycled; write=4.341 s, sync=0.026 s, total=4.417 s; sync files=12, longest=0.016 s, average=0.003 s; distance=318 kB, estimate=318 kB; lsn=0/1BEF630, redo lsn=0/1BEF508

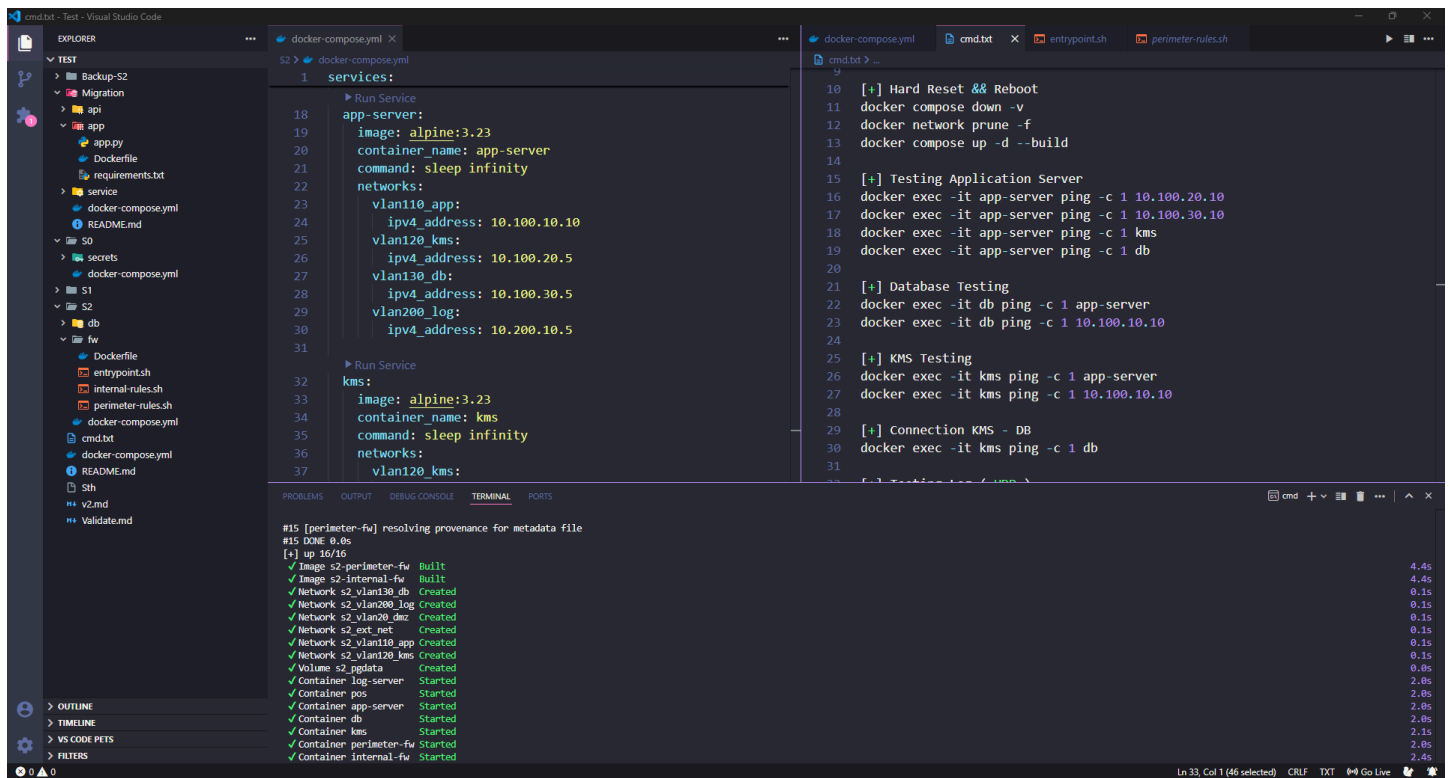
log-server

test log event

> 2026/06/06 17:38:12.000474253 length=15 from=0 to=14

test log event

(Container's Components Without Departments)



(Code and Testing Validation Sample Commands)

```

C:\Users\t0fu>docker exec -it app-server ping -c 1 kms
PING kms (10.100.20.10): 56 data bytes
64 bytes from 10.100.20.10: seq=0 ttl=64 time=0.244 ms

```

```

--- kms ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.244/0.244/0.244 ms

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug app-server](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```

C:\Users\t0fu>docker exec -it db ping -c 1 app-server
OCI runtime exec failed: exec failed: unable to start container process: exec: "ping": executable file not found in $PATH

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug db](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```

C:\Users\t0fu>docker exec -it kms ping -c 1 app-server
PING app-server (10.100.20.5): 56 data bytes
64 bytes from 10.100.20.5: seq=0 ttl=64 time=0.083 ms

```

```

--- app-server ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.083/0.083/0.083 ms

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug kms](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```

C:\Users\t0fu>docker exec -it kms ping -c 1 db
PING db (10.200.10.7): 56 data bytes
64 bytes from 10.200.10.7: seq=0 ttl=64 time=0.326 ms

```

```

--- db ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.326/0.326/0.326 ms

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug kms](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

(Validation on CDE)

2. Applicability Compliance Lists

Applicability Compliance Topic: Network Segmentation Effectiveness for PCI DSS Scope Reduction Name: THY DAYUTH		Summary of Applicable Requirements <table><tr><th>Requirement</th><th>Sub ID(s)</th><th>Description</th></tr><tr><td rowspan="5">Requirement 1</td><td>1.2.1</td><td>Plan secure network design, allowing all CDE connections</td></tr><tr><td>1.2.4</td><td>Maintain accurate data flow diagrams for network data</td></tr><tr><td>1.2.5</td><td>Identify and approve all allowed services, protocols, and ports</td></tr><tr><td>1.2.6</td><td>Implement security features for all allowed services and protocols</td></tr><tr><td>1.2.8</td><td>Secure configuration files from unauthorized access</td></tr><tr><td rowspan="5">Requirement 2</td><td>2.1</td><td>Restrict inbound traffic to the CDE to only what is necessary</td></tr><tr><td>2.2</td><td>Restrict outbound traffic from the CDE to only what is necessary</td></tr><tr><td>2.3</td><td>Implement NSGs between trusted and untrusted networks</td></tr><tr><td>2.4</td><td>Use automated access control systems for all components</td></tr><tr><td>2.5</td><td>Configure access systems to enforce job-based permissions</td></tr></table>		Requirement	Sub ID(s)	Description	Requirement 1	1.2.1	Plan secure network design, allowing all CDE connections	1.2.4	Maintain accurate data flow diagrams for network data	1.2.5	Identify and approve all allowed services, protocols, and ports	1.2.6	Implement security features for all allowed services and protocols	1.2.8	Secure configuration files from unauthorized access	Requirement 2	2.1	Restrict inbound traffic to the CDE to only what is necessary	2.2	Restrict outbound traffic from the CDE to only what is necessary	2.3	Implement NSGs between trusted and untrusted networks	2.4	Use automated access control systems for all components	2.5	Configure access systems to enforce job-based permissions	<table><tr><td rowspan="5">Requirement 1</td><td>1.2.1</td><td>Prohibit storage of Sensitive Authentication Data (SAD) post-auth</td></tr><tr><td>1.2.4</td><td>Mask SAD when displayed (omit BIN and last 6 digits)</td></tr><tr><td>1.2.5</td><td>Render stored SAD unreadable regardless of its access</td></tr><tr><td>1.2.6</td><td>Document the cryptographic architecture (Data in Transit)</td></tr><tr><td>1.2.7</td><td>Encrypt MFA with strong cryptography to non-public networks</td></tr><tr><td rowspan="5">Requirement 4</td><td>4.1</td><td>Implement strong cryptography for webserver authentication</td></tr><tr><td>4.2.1</td><td>Implement strong cryptography for webserver authentication</td></tr><tr><td>4.2.2</td><td>Implement strong cryptography for webserver authentication</td></tr><tr><td>4.2.3</td><td>Implement strong cryptography for webserver authentication</td></tr><tr><td>4.2.4</td><td>Implement strong cryptography for webserver authentication</td></tr><tr><td rowspan="5">Requirement 5</td><td>5.1</td><td>Design and implement an all-systems component</td></tr><tr><td>5.2</td><td>Design and implement an all-systems component</td></tr><tr><td>5.3</td><td>Design and implement an all-systems component</td></tr><tr><td>5.4</td><td>Design and implement an all-systems component</td></tr><tr><td>5.5</td><td>Design and implement an all-systems component</td></tr></table>	Requirement 1	1.2.1	Prohibit storage of Sensitive Authentication Data (SAD) post-auth	1.2.4	Mask SAD when displayed (omit BIN and last 6 digits)	1.2.5	Render stored SAD unreadable regardless of its access	1.2.6	Document the cryptographic architecture (Data in Transit)	1.2.7	Encrypt MFA with strong cryptography to non-public networks	Requirement 4	4.1	Implement strong cryptography for webserver authentication	4.2.1	Implement strong cryptography for webserver authentication	4.2.2	Implement strong cryptography for webserver authentication	4.2.3	Implement strong cryptography for webserver authentication	4.2.4	Implement strong cryptography for webserver authentication	Requirement 5	5.1	Design and implement an all-systems component	5.2	Design and implement an all-systems component	5.3	Design and implement an all-systems component	5.4	Design and implement an all-systems component	5.5	Design and implement an all-systems component																																																			
Requirement	Sub ID(s)	Description																																																																																																															
Requirement 1	1.2.1	Plan secure network design, allowing all CDE connections																																																																																																															
	1.2.4	Maintain accurate data flow diagrams for network data																																																																																																															
	1.2.5	Identify and approve all allowed services, protocols, and ports																																																																																																															
	1.2.6	Implement security features for all allowed services and protocols																																																																																																															
	1.2.8	Secure configuration files from unauthorized access																																																																																																															
Requirement 2	2.1	Restrict inbound traffic to the CDE to only what is necessary																																																																																																															
	2.2	Restrict outbound traffic from the CDE to only what is necessary																																																																																																															
	2.3	Implement NSGs between trusted and untrusted networks																																																																																																															
	2.4	Use automated access control systems for all components																																																																																																															
	2.5	Configure access systems to enforce job-based permissions																																																																																																															
Requirement 1	1.2.1	Prohibit storage of Sensitive Authentication Data (SAD) post-auth																																																																																																															
	1.2.4	Mask SAD when displayed (omit BIN and last 6 digits)																																																																																																															
	1.2.5	Render stored SAD unreadable regardless of its access																																																																																																															
	1.2.6	Document the cryptographic architecture (Data in Transit)																																																																																																															
	1.2.7	Encrypt MFA with strong cryptography to non-public networks																																																																																																															
Requirement 4	4.1	Implement strong cryptography for webserver authentication																																																																																																															
	4.2.1	Implement strong cryptography for webserver authentication																																																																																																															
	4.2.2	Implement strong cryptography for webserver authentication																																																																																																															
	4.2.3	Implement strong cryptography for webserver authentication																																																																																																															
	4.2.4	Implement strong cryptography for webserver authentication																																																																																																															
Requirement 5	5.1	Design and implement an all-systems component																																																																																																															
	5.2	Design and implement an all-systems component																																																																																																															
	5.3	Design and implement an all-systems component																																																																																																															
	5.4	Design and implement an all-systems component																																																																																																															
	5.5	Design and implement an all-systems component																																																																																																															
<table><tr><td rowspan="10">Requirement 6</td><td>6.1</td><td>Configure anti-malware for current endpoint updates</td></tr><tr><td>6.2</td><td>Automatically scan all removable electronic media</td></tr><tr><td>6.3</td><td>Enable anti-malware audits logs and retain them</td></tr><tr><td>6.4</td><td>Prevent users from disabling anti-malware mechanisms</td></tr><tr><td>6.5</td><td>Implement advanced technical protection against phishing</td></tr><tr><td>6.6</td><td>Develop bespoke software based on industry security standards</td></tr><tr><td>6.7</td><td>Perform pre-release code review for all custom software</td></tr><tr><td>6.8</td><td>Address common software attacks (e.g., injection, XSS)</td></tr><tr><td>6.9</td><td>Identify, rank, and manage security vulnerabilities</td></tr><tr><td>6.10</td><td>Identify, rank, and manage security vulnerabilities</td></tr></table>		Requirement 6	6.1	Configure anti-malware for current endpoint updates	6.2	Automatically scan all removable electronic media	6.3	Enable anti-malware audits logs and retain them	6.4	Prevent users from disabling anti-malware mechanisms	6.5	Implement advanced technical protection against phishing	6.6	Develop bespoke software based on industry security standards	6.7	Perform pre-release code review for all custom software	6.8	Address common software attacks (e.g., injection, XSS)	6.9	Identify, rank, and manage security vulnerabilities	6.10	Identify, rank, and manage security vulnerabilities	<table><tr><td rowspan="10">Requirement 7</td><td>7.1</td><td>Install critical security patches within one month of release</td></tr><tr><td>7.2</td><td>Protect public-facing web applications from attacks</td></tr><tr><td>7.3</td><td>Exclude pre-production and production environments</td></tr><tr><td>7.4</td><td>Exclude test/development and production environments</td></tr><tr><td>7.5</td><td>Remove test data/accounts before performing go-live</td></tr><tr><td>7.6</td><td>Define an access model based on "least privilege"</td></tr><tr><td>7.7</td><td>Assign access to system components and business needs to roles</td></tr><tr><td>7.8</td><td>Manage service accounts based on least privilege</td></tr><tr><td>7.9</td><td>Restrict user data queries to programmatic methods only</td></tr><tr><td>7.10</td><td>Restrict user data queries to programmatic methods only</td></tr></table>		Requirement 7	7.1	Install critical security patches within one month of release	7.2	Protect public-facing web applications from attacks	7.3	Exclude pre-production and production environments	7.4	Exclude test/development and production environments	7.5	Remove test data/accounts before performing go-live	7.6	Define an access model based on "least privilege"	7.7	Assign access to system components and business needs to roles	7.8	Manage service accounts based on least privilege	7.9	Restrict user data queries to programmatic methods only	7.10	Restrict user data queries to programmatic methods only	<table><tr><td rowspan="10">Requirement 8</td><td>8.1</td><td>Use automated access control systems for all components</td></tr><tr><td>8.2</td><td>Configure access systems to enforce job-based permissions</td></tr><tr><td>8.3</td><td>Set access control systems to "deny all" by default</td></tr><tr><td>8.4</td><td>Assign a unique ID to every user for all access</td></tr><tr><td>8.5</td><td>Immediately revoke access for terminated users</td></tr><tr><td>8.6</td><td>Remove or disable inactive user accounts within 90 days</td></tr><tr><td>8.7</td><td>Require re-authentication for idle sessions (15 mins)</td></tr><tr><td>8.8</td><td>Use at least one strong factor (password, token, biometric)</td></tr><tr><td>8.9</td><td>Render authentication factors unusable during transmission</td></tr><tr><td>8.10</td><td>Render authentication factors unusable during transmission</td></tr></table>		Requirement 8	8.1	Use automated access control systems for all components	8.2	Configure access systems to enforce job-based permissions	8.3	Set access control systems to "deny all" by default	8.4	Assign a unique ID to every user for all access	8.5	Immediately revoke access for terminated users	8.6	Remove or disable inactive user accounts within 90 days	8.7	Require re-authentication for idle sessions (15 mins)	8.8	Use at least one strong factor (password, token, biometric)	8.9	Render authentication factors unusable during transmission	8.10	Render authentication factors unusable during transmission	<table><tr><td rowspan="10">Requirement 9</td><td>9.1</td><td>Plan change of temporary passwords on first use</td></tr><tr><td>9.2</td><td>Enforce password complexity (min 12 chars, alphanumeric)</td></tr><tr><td>9.3</td><td>Prohibit reuse of the last four passwords</td></tr><tr><td>9.4</td><td>Implement MFA for non-console CDE administrative access</td></tr><tr><td>9.5</td><td>Implement MFA for all console CDE administrative access</td></tr><tr><td>9.6</td><td>Implement MFA for all external network access</td></tr><tr><td>9.7</td><td>Prohibit and justify interactive use of service accounts</td></tr><tr><td>9.8</td><td>Prohibit hard-coded passwords in scripts or source code</td></tr><tr><td>9.9</td><td>Prohibit hard-coded passwords in scripts or source code</td></tr><tr><td>9.10</td><td>Prohibit hard-coded passwords in scripts or source code</td></tr></table>		Requirement 9	9.1	Plan change of temporary passwords on first use	9.2	Enforce password complexity (min 12 chars, alphanumeric)	9.3	Prohibit reuse of the last four passwords	9.4	Implement MFA for non-console CDE administrative access	9.5	Implement MFA for all console CDE administrative access	9.6	Implement MFA for all external network access	9.7	Prohibit and justify interactive use of service accounts	9.8	Prohibit hard-coded passwords in scripts or source code	9.9	Prohibit hard-coded passwords in scripts or source code	9.10	Prohibit hard-coded passwords in scripts or source code	<table><tr><td rowspan="10">Requirement 10</td><td>10.1.1</td><td>Log all individual user access to confidential data</td></tr><tr><td>10.1.2</td><td>Log all creation and deletion of system-level objects</td></tr><tr><td>10.1.3</td><td>Capture specific details for every audit log entry</td></tr><tr><td>10.1.4</td><td>Logging must occur in close with a business need</td></tr><tr><td>10.1.5</td><td>Protect audit logs from unauthorized modifications</td></tr><tr><td>10.1.6</td><td>Back up logs to a secure, non-volatile log store</td></tr><tr><td>10.1.7</td><td>Address and remediate log addresses and anomalies</td></tr><tr><td>10.1.8</td><td>Synchronize all system clocks using NTP technology</td></tr><tr><td>10.1.9</td><td>Obtain and obtain on behalf of individual security controls</td></tr><tr><td>10.1.10</td><td>Obtain and obtain on behalf of individual security controls</td></tr></table>	Requirement 10	10.1.1	Log all individual user access to confidential data	10.1.2	Log all creation and deletion of system-level objects	10.1.3	Capture specific details for every audit log entry	10.1.4	Logging must occur in close with a business need	10.1.5	Protect audit logs from unauthorized modifications	10.1.6	Back up logs to a secure, non-volatile log store	10.1.7	Address and remediate log addresses and anomalies	10.1.8	Synchronize all system clocks using NTP technology	10.1.9	Obtain and obtain on behalf of individual security controls	10.1.10	Obtain and obtain on behalf of individual security controls
Requirement 6	6.1		Configure anti-malware for current endpoint updates																																																																																																														
	6.2		Automatically scan all removable electronic media																																																																																																														
	6.3		Enable anti-malware audits logs and retain them																																																																																																														
	6.4		Prevent users from disabling anti-malware mechanisms																																																																																																														
	6.5		Implement advanced technical protection against phishing																																																																																																														
	6.6		Develop bespoke software based on industry security standards																																																																																																														
	6.7		Perform pre-release code review for all custom software																																																																																																														
	6.8		Address common software attacks (e.g., injection, XSS)																																																																																																														
	6.9		Identify, rank, and manage security vulnerabilities																																																																																																														
	6.10	Identify, rank, and manage security vulnerabilities																																																																																																															
Requirement 7	7.1	Install critical security patches within one month of release																																																																																																															
	7.2	Protect public-facing web applications from attacks																																																																																																															
	7.3	Exclude pre-production and production environments																																																																																																															
	7.4	Exclude test/development and production environments																																																																																																															
	7.5	Remove test data/accounts before performing go-live																																																																																																															
	7.6	Define an access model based on "least privilege"																																																																																																															
	7.7	Assign access to system components and business needs to roles																																																																																																															
	7.8	Manage service accounts based on least privilege																																																																																																															
	7.9	Restrict user data queries to programmatic methods only																																																																																																															
	7.10	Restrict user data queries to programmatic methods only																																																																																																															
Requirement 8	8.1	Use automated access control systems for all components																																																																																																															
	8.2	Configure access systems to enforce job-based permissions																																																																																																															
	8.3	Set access control systems to "deny all" by default																																																																																																															
	8.4	Assign a unique ID to every user for all access																																																																																																															
	8.5	Immediately revoke access for terminated users																																																																																																															
	8.6	Remove or disable inactive user accounts within 90 days																																																																																																															
	8.7	Require re-authentication for idle sessions (15 mins)																																																																																																															
	8.8	Use at least one strong factor (password, token, biometric)																																																																																																															
	8.9	Render authentication factors unusable during transmission																																																																																																															
	8.10	Render authentication factors unusable during transmission																																																																																																															
Requirement 9	9.1	Plan change of temporary passwords on first use																																																																																																															
	9.2	Enforce password complexity (min 12 chars, alphanumeric)																																																																																																															
	9.3	Prohibit reuse of the last four passwords																																																																																																															
	9.4	Implement MFA for non-console CDE administrative access																																																																																																															
	9.5	Implement MFA for all console CDE administrative access																																																																																																															
	9.6	Implement MFA for all external network access																																																																																																															
	9.7	Prohibit and justify interactive use of service accounts																																																																																																															
	9.8	Prohibit hard-coded passwords in scripts or source code																																																																																																															
	9.9	Prohibit hard-coded passwords in scripts or source code																																																																																																															
	9.10	Prohibit hard-coded passwords in scripts or source code																																																																																																															
Requirement 10	10.1.1	Log all individual user access to confidential data																																																																																																															
	10.1.2	Log all creation and deletion of system-level objects																																																																																																															
	10.1.3	Capture specific details for every audit log entry																																																																																																															
	10.1.4	Logging must occur in close with a business need																																																																																																															
	10.1.5	Protect audit logs from unauthorized modifications																																																																																																															
	10.1.6	Back up logs to a secure, non-volatile log store																																																																																																															
	10.1.7	Address and remediate log addresses and anomalies																																																																																																															
	10.1.8	Synchronize all system clocks using NTP technology																																																																																																															
	10.1.9	Obtain and obtain on behalf of individual security controls																																																																																																															
	10.1.10	Obtain and obtain on behalf of individual security controls																																																																																																															

As for right now I have established the baseline of what I’m choosing, apologized for not organizing it better before submission but it’s to have the professor review the draft first.

Note : Some requirements might be add or remove in the future 🙏

3. Time Planning Table : [Link](#)

Project Management Plan

PROJECT TITLE: Network Segmentation Effectiveness for PCI DSS Scope

STUDENT: Mr. Thy Dayuth

ADVISOR: Mr. Meas Sotheareth

COMPANY NAME: ACLEDA Bank Plc.

PROJECT SUPERVISOR: Mr. Phok Visal

WBS NUMBER	TASK TITLE	START DATE	COMPLETED DATE	DURATION	TASK STATUS	PHASE TWO		PHASE THREE		PHASE FOUR		PHASE FIVE		PHASE SIX	
						WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11	WEEK 12			
						M	T	W	R	F	M	T	W	R	F
Category A : Core Network & Infrastructure															
A01	Configure Switch 1 Dept VLANs	25-May-26	27-May-26	2	COMPLETED										
A02	Configure Switch 2 CDE Segmentation	27-Dec-99	29-Dec-99	2	COMPLETED										
A03	Configure Inbound/Outbound Firewall Rules	29-May-26	31-May-26	2	COMPLETED										
A04	Provision Perimeter FW (VLAN 10/20)	31-May-26	01-Jun-26	1	COMPLETED										
A05	Provision Internal FW (CDE & Logs)	01-Jun-26	03-Jun-26	2	COMPLETED										
Category B : Security Controls & Firewall Rules															
B01	Implement Department Isolation ACLs	03-Jun-26	05-Jun-26	2	COMPLETED										
B02	Configure Switch 1 Jumphost Routing			0	IN PROGRESS										
B03	Establish Retailer-to-DMZ TLS 1.2 Tunnel			0	IN PROGRESS										
B04	Restrict App Server KMS/DB Traffic	05-Jun-26	06-Jun-26	1	COMPLETED										
B05	Enforce Deny-by-Default (FW & Switch 1)	06-Jun-26	09-Jun-26	3	COMPLETED										
Category C : Systems & Identity Access Management (IAM)															
C01	Deploy Jumphost (Static IP)			0	NOT STARTED										
C02	Harden Jumphost Server			0	NOT STARTED										
C03	Implement Jumphost MFA			0	NOT STARTED										
C04	Deploy Logging Server Pipeline	04-Jun-26	07-Jun-26	3	IN PROGRESS										
Category D : Database & Application Development															
D01	Provision PostgreSQL & DB Roles			0	IN PROGRESS										
D02	Initialize DB Schema			0	IN PROGRESS										
D03	Integrate Key Management System (KMS)			0	IN PROGRESS										
D04	Develop App Server Inbound Handling			0	IN PROGRESS										
D05	Integrate App Server KMS API			0	IN PROGRESS										
D06	Develop App Server DB Persistence Layer			0	IN PROGRESS										
D07	Develop POS Payload Generation Script			0	IN PROGRESS										
Category E : Integration, Testing & End-to-End Validation															
E01	Conduct Network Isolation Nmap Scan			0	NOT STARTED										
E02	Verify MFA Integration			0	NOT STARTED										
E03	Verify CDE & KMS/DB Access Controls			0	NOT STARTED										
E04	Simulate End-to-End POS Transaction			0	NOT STARTED										
E05	Validate Security Audit Logs			0	NOT STARTED										
Category F : Documentation & Reporting (Final Phase)															
F01	Finalize Project Report			7 days	NOT STARTED										
					NOT STARTED										

As for now, I have completed some of the core which is the structure both starting on the CDE code as well, I’ll keep it up to date as much as possible.

Note : Some are planning, Some finished for a phase but still need monitoring in case there’s modification